VNR Vignana Jyothi Institute of Engineering

Electronics and Communication Engineering | Course Code: EC305

Communication Systems & Modulation Protocols - Laboratory Manual & Pract

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Syllabus Focus: EC305 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Amplitude Modulation & Detection

" Syllabus Scope: Double Sideband Full Carrier DSB-FC, DSB-SC, Single Sideband SSB-SC, Vestigial Sideband

Unit 2: Frequency & Phase Modulation

" Syllabus Scope: Narrowband and wideband FM, Carsons bandwidth rule, indirect FM Armstrong method, phase

EC305: Communication Systems & Modulation Protocols - Advanced Topics & Assessme

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Pulse Modulation & Baseband Transmission

" Sampling theorem, PAM, PWM, PPM, Pulse Code Modulation PCM, uniform and non-uniform quantization A-law, mu-l

Unit 4: Digital Bandpass Modulation Schemes

" Geometric signal representation, Binary Phase Shift Keying BPSK, QPSK, M-ary QAM, constellation diagrams, ma

Unit 5: Noise Analysis & Information Theory

" Thermal noise, SNR in AM/FM receivers, Shannon channel capacity theorem, entropy, source coding Huffman, Sha

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Generation and Demodulation of Amplitude Modulated AM Waveform

Lab 2: Frequency Modulation FM and Demodulation using Phase-Locked Loop PLL

Lab 3: Pulse Code Modulation PCM and Demodulation with Quantization Noise Measurement

Lab 4: Digital Modulation and Constellation Diagram Verification for BPSK and QPSK

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Simon Haykin - Communication Systems Wiley 4th Ed

* B.P. Lathi, Zhi Ding - Modern Digital and Analog Communication Systems Oxford University Press 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students